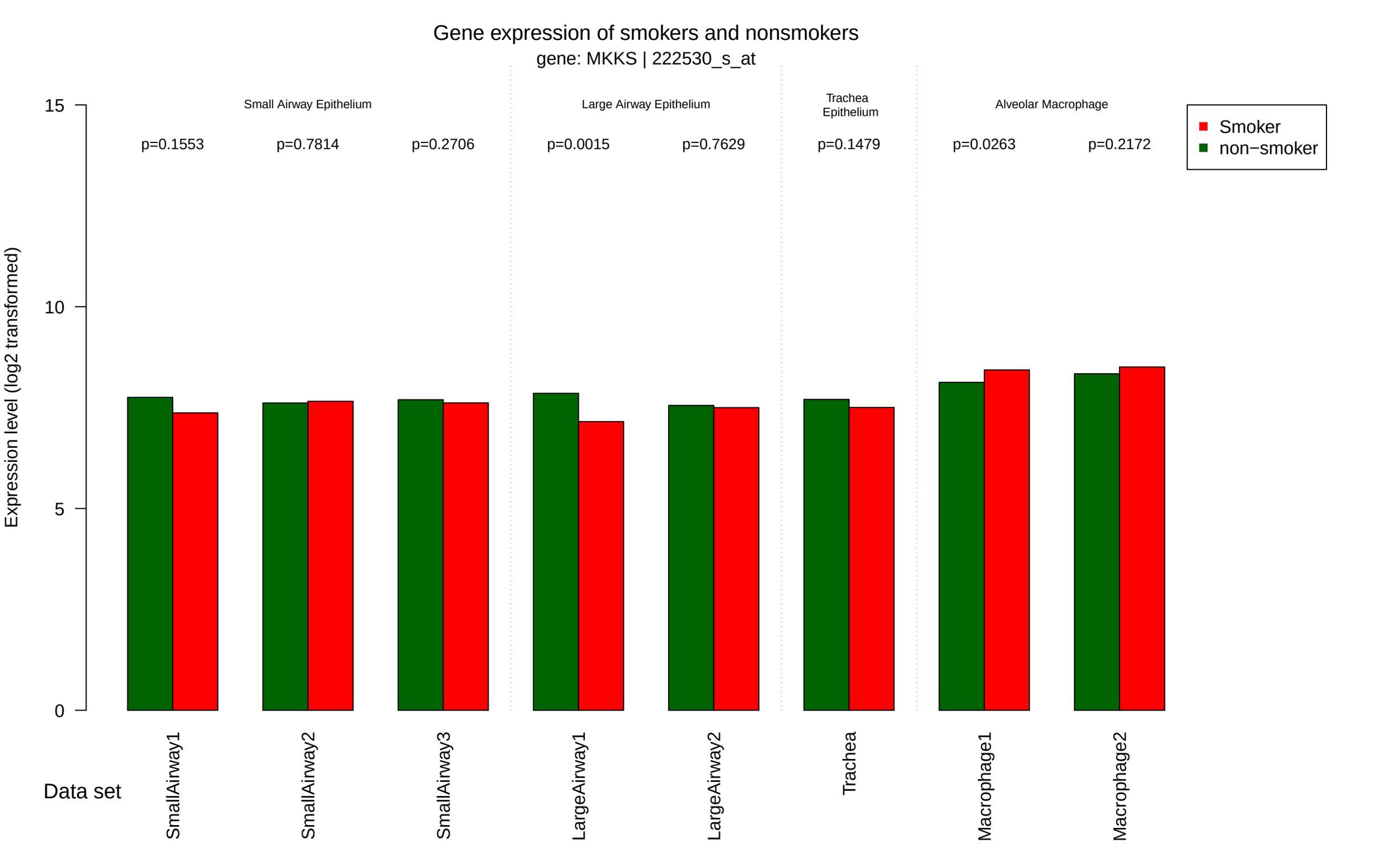
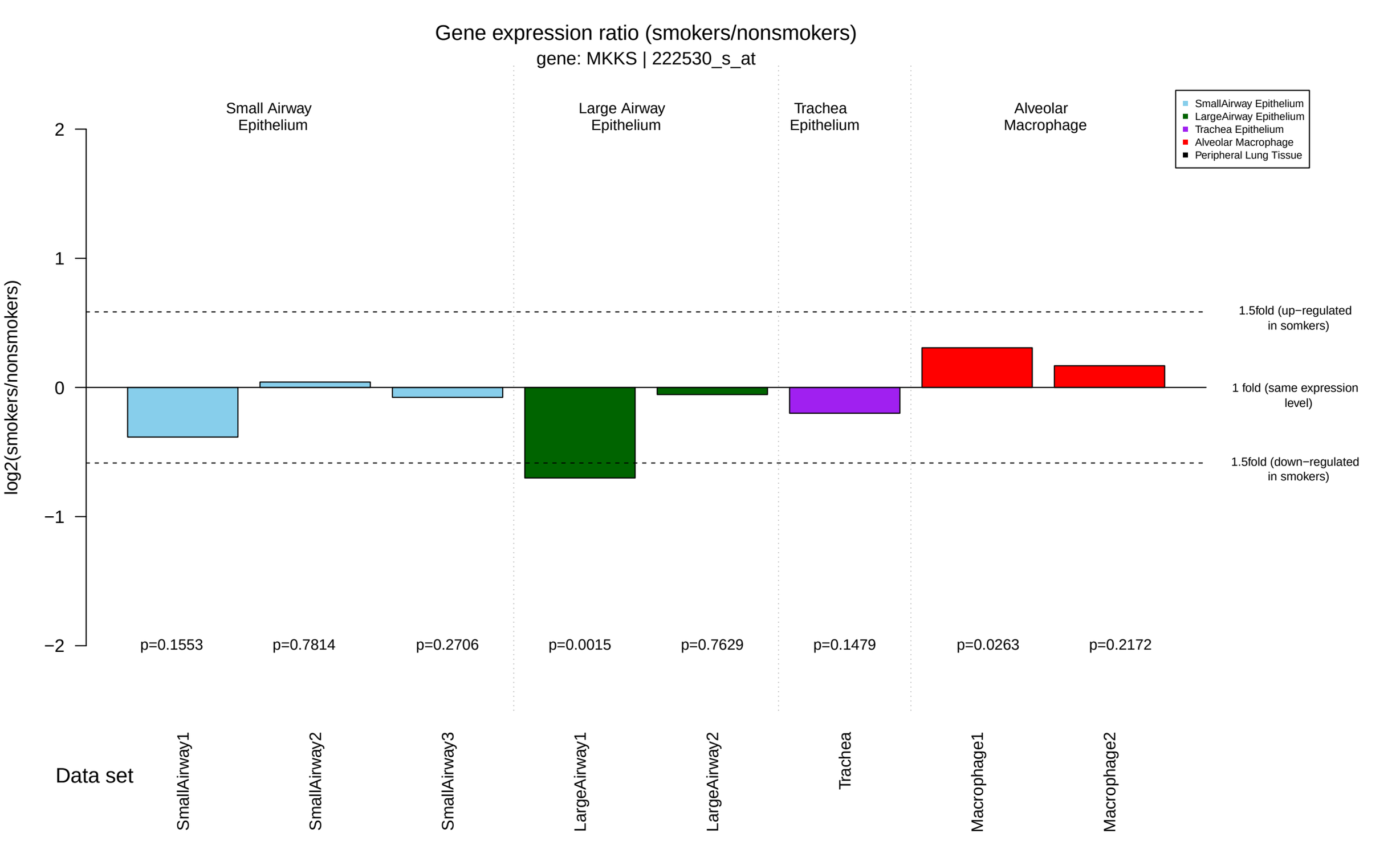
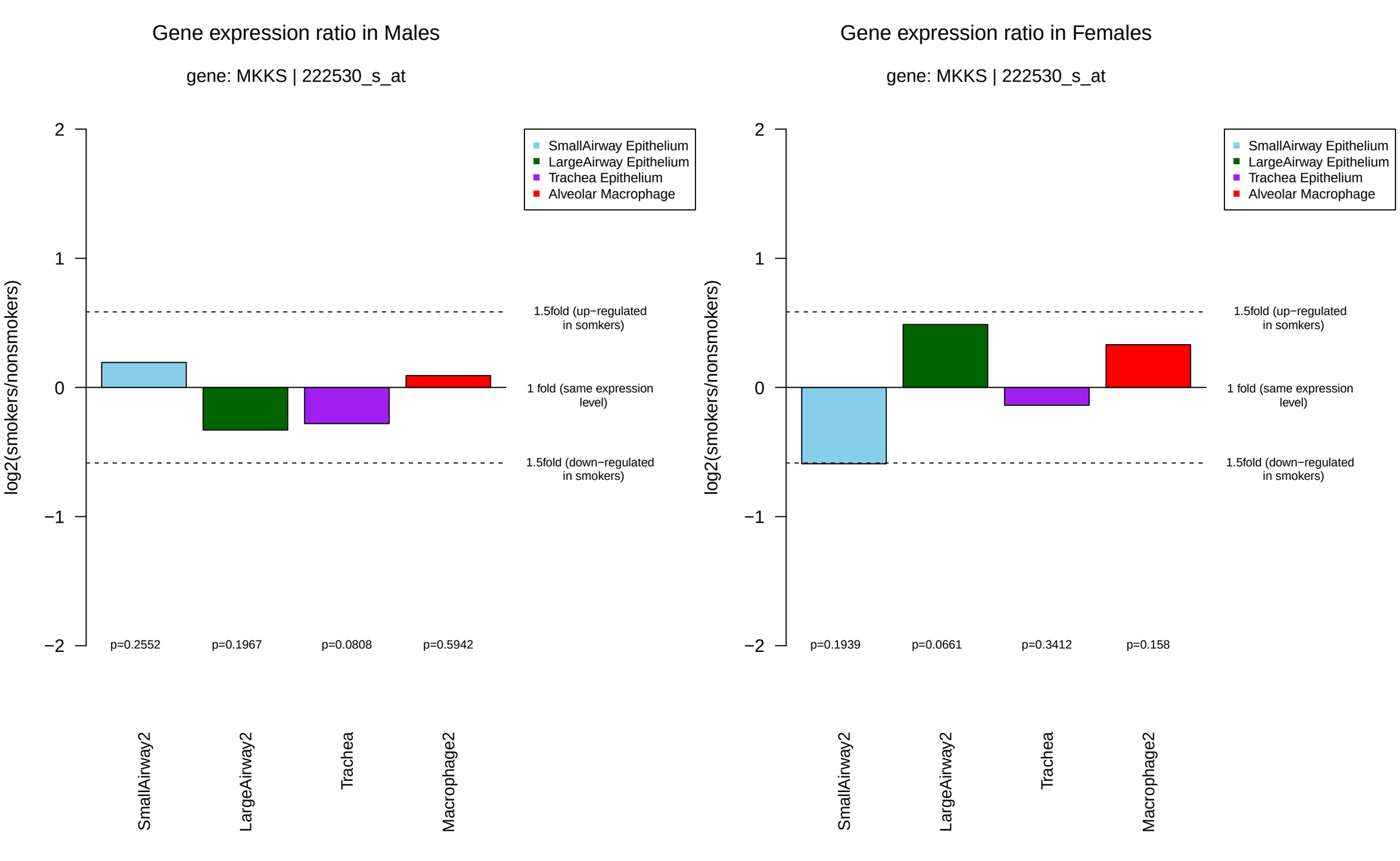
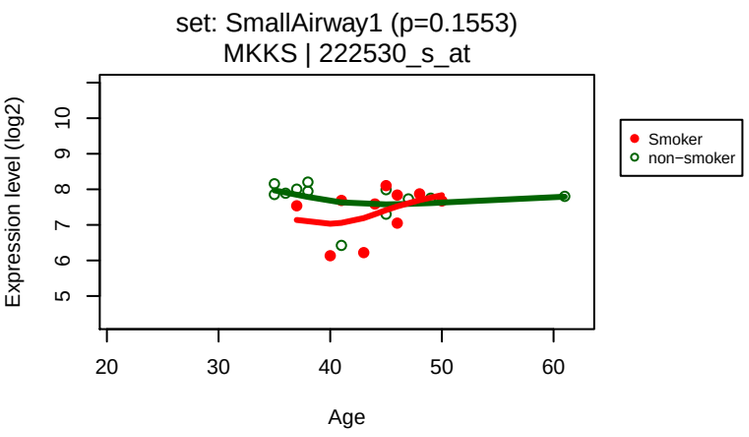
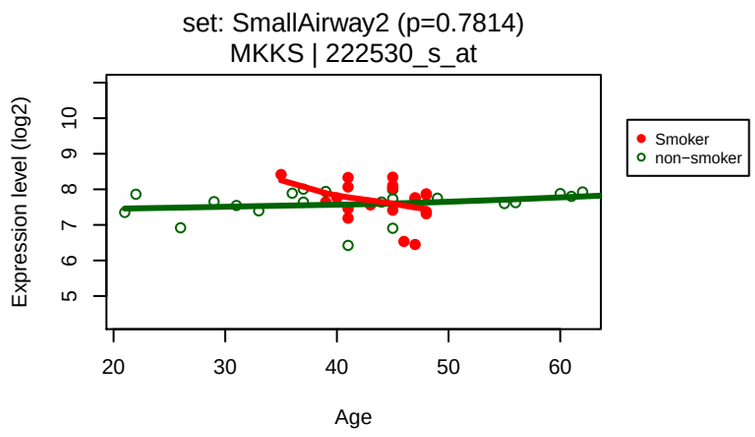
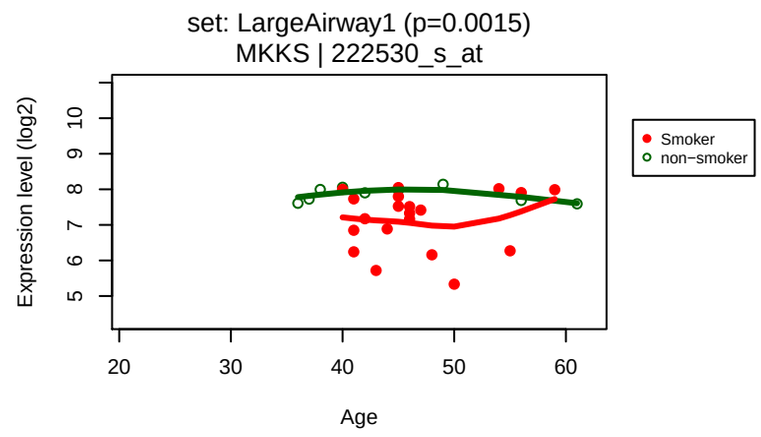

Gene expression of smokers and nonsmokers
gene: MKKS | 222530_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1553
p=0.7814
p=0.2706
p=0.0015
p=0.7629
p=0.1479
p=0.0263
p=0.2172
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

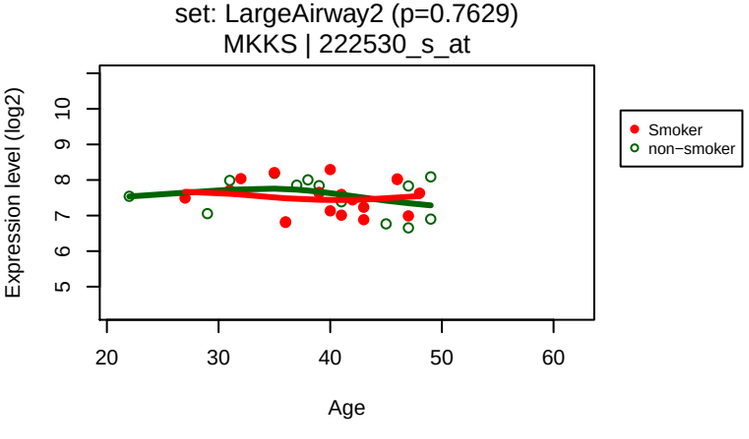

Gene expression ratio (smokers/nonsmokers)
gene: MKKS | 222530_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1553
p=0.7814
p=0.2706
p=0.0015
p=0.7629
p=0.1479
p=0.0263
p=0.2172
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

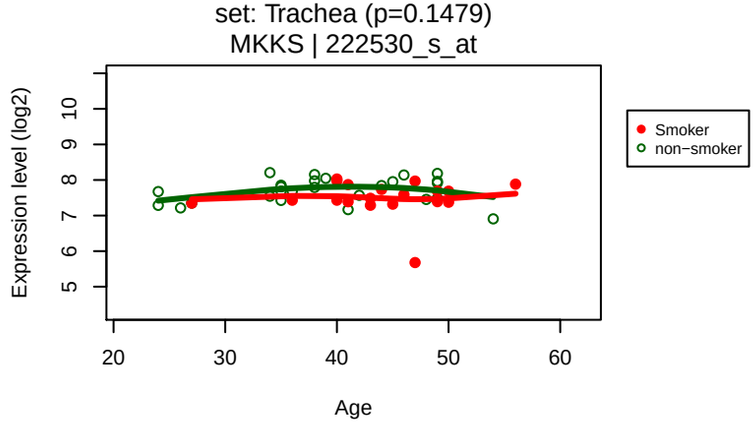

Gene expression ratio in Males
gene: MKKS | 222530_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2552
p=0.1967
p=0.0808
p=0.5942
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MKKS | 222530_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1939
p=0.0661
p=0.3412
p=0.158
SmallAirway2
LargeAirway2
Trachea
Macrophage2

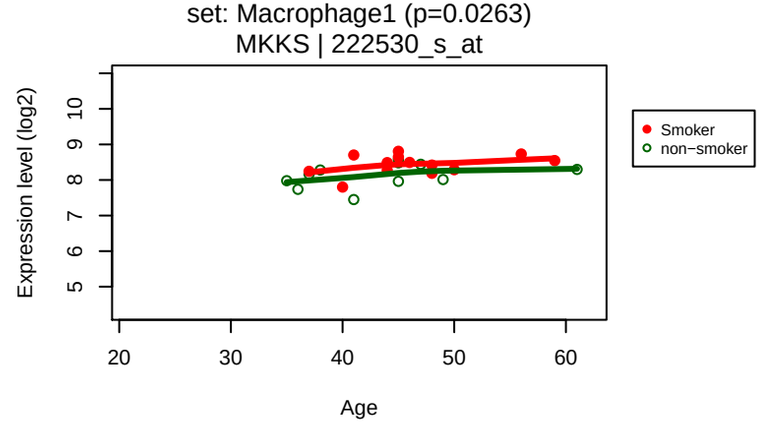

set: SmallAirway1 (p=0.1553)
MKKS | 222530_s_at
Expression level (log2)
Age
Smoker
non−smoker

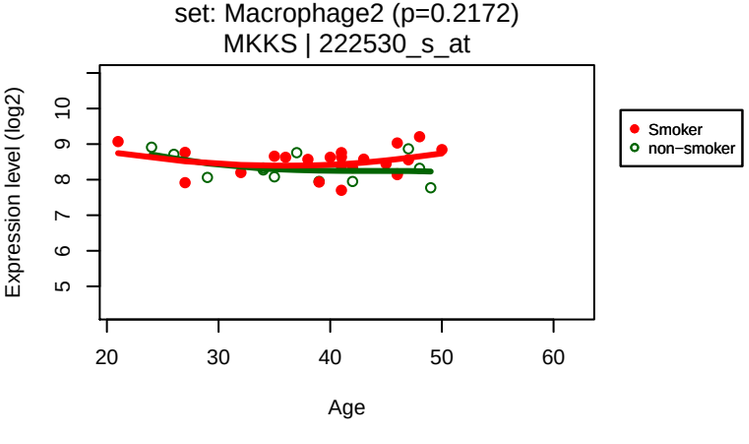

set: SmallAirway2 (p=0.7814)
MKKS | 222530_s_at
Expression level (log2)
Age
Smoker
non−smoker

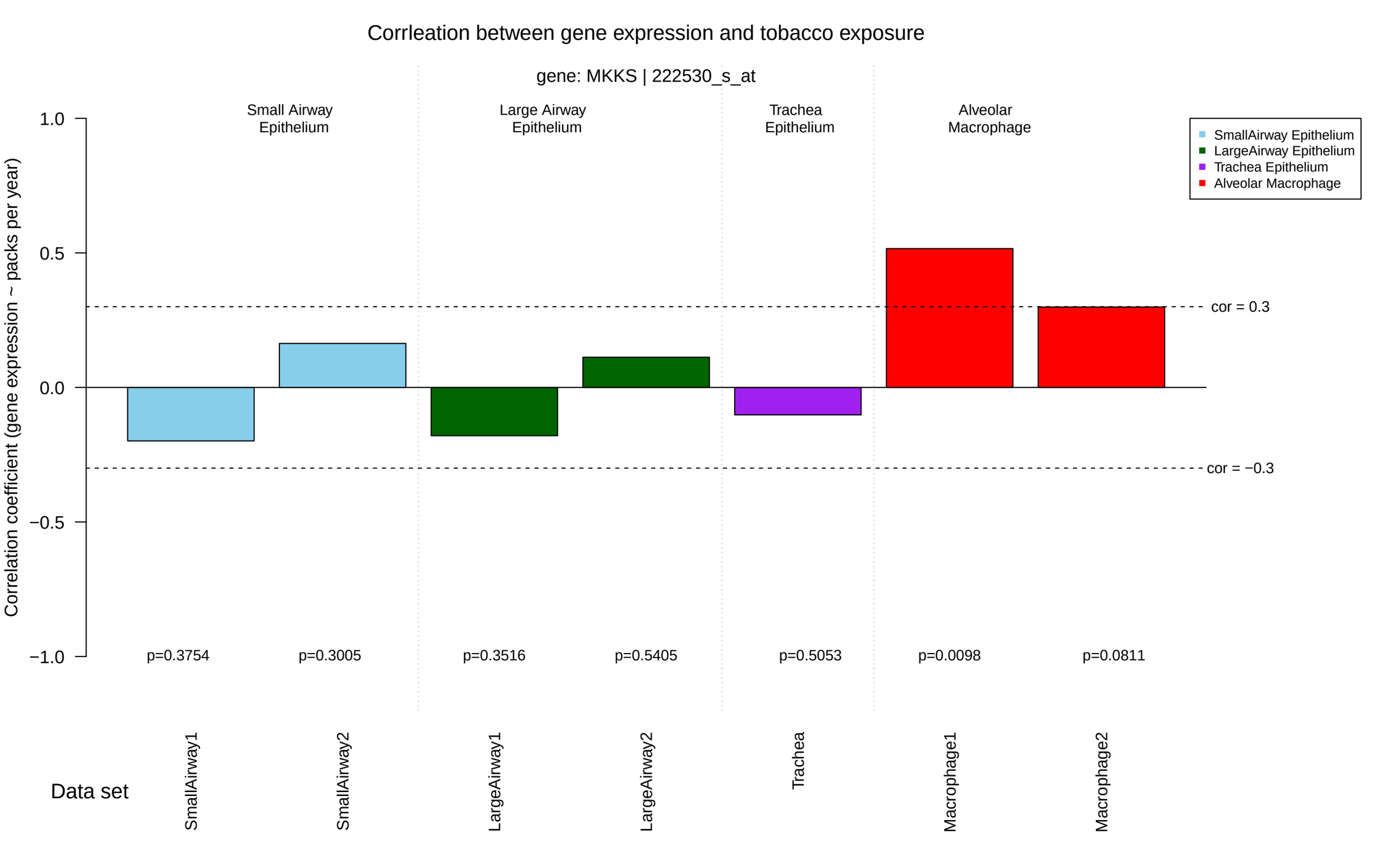

set: LargeAirway1 (p=0.0015)
MKKS | 222530_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7629)
MKKS | 222530_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1479)
MKKS | 222530_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0263)
MKKS | 222530_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2172)
MKKS | 222530_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MKKS | 222530_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3754
p=0.3005
p=0.3516
p=0.5405
p=0.5053
p=0.0098
p=0.0811
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2